



DATA ANALYSIS REPORT

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PROJECT OVERVIEW

CONTEXT

The aim of this project is to analyse hospital performance data and generate insights to support data-driven decision-making. The analysis focuses on key areas including patient behaviour, encounter patterns, cost and coverage trends, and readmission rates.

DATASET

The dataset for this project comes from Maven Analytics and consists of synthetic data representing approximately 1,000 patients from Massachusetts General Hospital over the period 2011–2022. It includes information on patient demographics, insurance coverage, and detailed records of medical encounters and procedures.

DATA PREPARATION

The database was reconstructed in PostgreSQL, and the data was imported and structured using SQL queries to support analysis.

SCOPE OF ANALYSIS

The scope of this analysis covers four key areas: patient demographics, patient utilization, encounter patterns, and cost and coverage trends.

SUMMARY OF FINDINGS

- The patient population is skewed toward older adults, with a large proportion aged 76+, who also account for the majority of healthcare utilisation.
- Healthcare utilisation is uneven across patients. While some patients have only a single visit, most are repeat users, and a small number of patients account for a very high number of encounters.
- Encounter activity is generally stable over time, with a few peaks in certain years. Most encounters are ambulatory or outpatient, and the vast majority are short in duration (less than 24 hours).
- Nearly half of all encounters have no payer coverage, indicating a high level of out-of-pocket healthcare usage. Routine procedures drive most of the volume, while a small number of complex procedures account for the highest costs.
- A large proportion of patients have follow-up visits within 30 days, but when focusing only on inpatient cases, the readmission rate is much lower. This shows that broader definitions can overestimate true readmission levels.

ANALYSIS

Patient Demographics

Objective

How are patients distributed across age groups, and which group accounts for the highest number of encounters?

Approach

Patients were grouped into five age categories (<18, 18–35, 36–55, 56–75, 76+), and counts were aggregated to determine the distribution. Encounter volumes were also calculated for each age group.

Query

```
-- creating age groups based on age at the time of the encounter and counting patient numbers
SELECT
  CASE
    WHEN EXTRACT(YEAR FROM AGE(e.start_time, birthdate)) < 18 THEN 'Under 18'
    WHEN EXTRACT(YEAR FROM AGE(e.start_time, birthdate)) BETWEEN 18 AND 35 THEN '18-35'
    WHEN EXTRACT(YEAR FROM AGE(e.start_time, birthdate)) BETWEEN 36 AND 55 THEN '36-55'
    WHEN EXTRACT(YEAR FROM AGE(e.start_time, birthdate)) BETWEEN 56 AND 75 THEN '56-75'
    ELSE '76+'
  END AS age_group,
  COUNT (DISTINCT p.id) AS patient_count,
  ROUND(COUNT (DISTINCT p.id) * 100.0 / SUM(COUNT (DISTINCT p.id)) OVER (), 1) as
percentage
FROM patients p
JOIN encounters e on
  p.id = e.patient_id
GROUP BY age_group
ORDER BY percentage DESC;
```

Results

- 33.5% of patients are aged 76+
- 55.7% of the total encounters are generated by this age group.

Key Findings

A large proportion of patients in the dataset are older adults, indicating that healthcare demand is largely driven by age-related conditions and the need for ongoing elderly care.

Patient Utilisation

Objective

How do patients utilise healthcare services and how is encounter frequency distributed across the patient population?

Approach

Patient-level encounter counts were calculated to analyse visit frequency, identify repeat patients, and highlight high-utilization individuals.

Query

a) Patient distribution by encounter

```
-- Patients to encounter distribution using sub query

SELECT
  encounter_count,
  COUNT(*) AS number_of_patients
```

```
FROM
(
    SELECT patient_id,
           COUNT(*) AS encounter_count
    FROM encounters
    GROUP BY patient_id
) t
GROUP BY encounter_count
ORDER BY encounter_count;
```

Results

- Single visits have the highest count (120).
- The number of patients decreases significantly after the first visit (50 for two visits, 40 for three visits).

b) Single vs Repeat Patients

```
-- Creating a temp table for patient data
CREATE TEMP TABLE patient_visits as
SELECT
    patient_id,
    count(*) as encounter_count
FROM
    encounters
GROUP BY
    patient_id;

-- checking single or multiple visits and their percentage
SELECT
    COUNT(*) FILTER (WHERE encounter_count = 1) AS single_visit_patients,
    ROUND(COUNT(*) FILTER (WHERE encounter_count = 1) * 100.0 / COUNT(*),1)
    AS single_visit_percentage,
    COUNT(*) FILTER (WHERE encounter_count > 1) AS repeat_patients,
    ROUND(COUNT(*) FILTER (WHERE encounter_count > 1) * 100.0 / COUNT(*),1)
    AS repeat_percentage
FROM patient_visits;
```

Result

- A high proportion of patients are repeat users (87.7%)

c) High-utilisation patients

```
SELECT *
from patient_visits
order by encounter_count DESC
limit 10;
```

Result

- The top three patients recorded 1,381, 887, and 877 encounters respectively.

Key Findings

Healthcare utilization is highly uneven across the patient population. While a large proportion of patients have only a small number of visits, the majority are repeat users of healthcare services. Additionally, a small subset of patients accounts for an exceptionally high number of encounters, indicating the presence of high-utilization individuals likely requiring ongoing or complex care.

Encounter Analysis

Objective

How are encounters distributed over time, which encounter classes are most common, and what proportion of encounters last less than 24 hours?

Approach

Encounter data was analyzed to examine trends over time, identify the most common encounter classes, and determine the proportion of short-duration encounters.

Query

a) Encounters by year

```
-- overall encounters by year
SELECT
    EXTRACT(YEAR FROM start_time) AS year,
    COUNT(*) AS total_encounters
FROM encounters
WHERE
    start_time < '2022-01-01'
GROUP BY year
ORDER BY year;
-- data for 2022 was excluded because it wasn't the complete year (only up till Feb)
```

Results

- Average encounter counts were above 2,000 for most years.
- 2014 and 2021 saw an unusual peak in encounters (3885, 3530 respectively)

b) Percentage encounter distribution by year and class

```
SELECT
    EXTRACT(YEAR FROM start_time) AS year,
    ROUND(COUNT(*) FILTER (WHERE encounterclass = 'ambulatory') * 100.0 / COUNT(*), 1) AS
    ambulatory,
    ROUND(COUNT(*) FILTER (WHERE encounterclass = 'outpatient') * 100.0 / COUNT(*), 1)
    AS outpatient,
    ROUND(COUNT(*) FILTER (WHERE encounterclass = 'inpatient') * 100.0 / COUNT(*), 1)
    AS inpatient,
    ROUND(COUNT(*) FILTER (WHERE encounterclass = 'emergency') * 100.0 / COUNT(*), 1)
    AS emergency,
    ROUND(COUNT(*) FILTER (WHERE encounterclass = 'urgentcare') * 100.0 / COUNT(*), 1) AS
    urgentcare,
    ROUND(COUNT(*) FILTER (WHERE encounterclass = 'wellness') * 100.0 / COUNT(*), 1) AS
    wellness
FROM encounters
GROUP BY year
ORDER BY year;
```

Results

- The highest percentage of encounters were ambulatory and outpatient

c) Percentage of encounters less than 24 hours

```

SELECT
CASE WHEN (end_time - start_time) < INTERVAL '24 hours' THEN 'under_24' ELSE 'over_24'
END AS duration_group,
COUNT(*) AS total_encounters,
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2 ) AS percentage
FROM encounters
GROUP BY duration_group;

```

Result

- 95.87% of all encounters where less than 24 hours

Key Findings

Encounter trends remain relatively stable over time, with significant spikes in 2014 and 2021 indicating periods of heightened healthcare demand. The analysis reveals a clear dominance of ambulatory and outpatient encounters, suggesting that healthcare utilization is primarily driven by routine and non-inpatient services. In addition, over 95% of encounters last less than 24 hours, highlighting that most healthcare interactions are short-term and non-intensive in nature.

COST AND COVERAGE ANALYSIS

Objective

How are healthcare costs distributed across payers, what proportion of encounters have no insurance coverage, and which procedures drive the highest cost and volume?

Approach

Cost and coverage data were analyzed to assess the proportion of uninsured encounters, compare average costs across payers, and identify procedures that drive both high volume and high cost.

Query

a) Zero coverage analysis

```

-- getting percentage of zero pay_coverage
SELECT
COUNT(*) AS ZERO_PERCENTAGE,
ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM encounters),2) as percentage_zero
FROM encounters
where payer_coverage = 0.00; --48.71 % of encounters have zero pay charge

```

Result

- 48.7% of encounters have zero payer coverage, indicating a high proportion of uninsured or out-of-pocket cases.

b) Average cost by payer

```

SELECT p.name, ROUND(AVG(e.total_claim_cost), 2) as average_cost
FROM
encounters e

```

```

        join payers p
        ON e.payer_id = p.id
GROUP BY p.name
ORDER BY average_cost DESC;

```

Result

- Medicaid has the highest average cost (\$6,205.22) followed by encounters with no insurance (\$5,593).

c) Procedures analysis

```

-- TOP 5 PROCEDURES BY AVERAGE COST
SELECT
    description,
    COUNT(*) AS procedure_count,
    ROUND(AVG(base_cost), 2) AS avg_cost
FROM procedures
GROUP BY description
ORDER BY avg_cost DESC
LIMIT 5;

--TOP 5 PROCEDURES BY QUANTITY PERFORMED
SELECT
    description,
    COUNT(*) as procedure_total,
    ROUND(AVG(base_cost),2) as avg_cost
FROM
    procedures
GROUP BY description
ORDER BY procedure_total desc
LIMIT 5;

```

Results

- High-cost procedures include ICU admissions and coronary artery bypass grafting.
- The most common procedures are routine procedures and screenings such as health evaluation and depression screening.

Key Findings

The analysis reveals that nearly half of all encounters have no payer coverage, indicating a substantial reliance on out-of-pocket payments. Cost patterns vary across payers, with higher average costs observed for public insurance and uninsured patients. Additionally, there is a clear separation between volume and cost drivers: routine procedures dominate in frequency, while a small number of complex interventions, including ICU admissions and major surgeries, account for a disproportionate share of total costs.

READMISSION

Objective

What percentage of patients are readmitted within 30 days and what does this indicate about continuity of care?

Approach

Encounter data was analysed to identify patients who were readmitted within 30 days of last encounter.

Query

a) Readmission across all patients

```
WITH pd as (  
    SELECT patient_id, start_time,  
           LEAD(start_time) over (partition by patient_id order by start_time)  
           As next_appointment  
    from encounters  
)  
  
readmissions as (  
    SELECT count(distinct patient_id) as readmin_patients  
    from pd  
    where next_appointment is not null AND next_appointment - start_time  
    < INTERVAL '30 days'),  
  
total_p as (  
    SELECT COUNT(DISTINCT patient_id) as total_patients from encounters)  
  
SELECT readmin_patients, total_patients,  
       ROUND(readmin_patients * 100 / total_patients,2) as percentage_readmission  
from readmissions, total_p;
```

Result

- 79% of patients were readmitted within 30 days

b) Readmission across inpatient class only

```
-- creating a CTE  
WITH inpatient_visits AS (  
    SELECT  
        patient_id, start_time  
    FROM encounters  
    WHERE encounterclass = 'inpatient'  
)  
  
patient_visits AS (  
    SELECT patient_id, start_time,  
           LEAD(start_time) OVER (PARTITION BY patient_id  
                                   ORDER BY start_time  
                                ) AS next_start_time  
    FROM inpatient_visits  
)  
  
readmission_flags AS (  
    SELECT  
        patient_id,  
        CASE  
            WHEN next_start_time IS NOT NULL  
            AND next_start_time - start_time <= INTERVAL '30 days'  
            THEN 1  
            ELSE 0  
        END AS is_readmission  
    FROM patient_visits
```

```

),

patient_level AS (
    SELECT
        patient_id,
        MAX(is_readmission) AS had_readmission
    FROM readmission_flags
    GROUP BY patient_id
)

SELECT
    COUNT(*) AS total_inpatient_patients,
    SUM(had_readmission) AS patients_with_readmission,
    ROUND(
        SUM(had_readmission) * 100.0 / COUNT(*),
        2
    ) AS readmission_rate_percentage
FROM patient_level;

```

Result

- 26.8% of inpatient encounters were readmissions

Key Findings

Although 79% of patients had a follow-up visit within 30 days, this includes routine and planned care. A more focused analysis of inpatient cases shows a lower rate of 26.8%, which better reflects meaningful readmissions. This highlights the importance of using clear definitions when interpreting results.